

ARJUN S NAIR

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PROFESSIONAL SUMMARY

Full-stack developer currently working as a Full Stack AI Integration Developer at IFOA (International Flight Operational Academy), where I build AI-integrated learning, certification, and operations platforms using the MERN stack and LLM APIs (Gemini, Groq). Computer Science undergraduate with a strong foundation in Python, REST API design, and production-grade authentication and payment systems. Patent holder and competitive hackathon qualifier.

TECHNICAL SKILLS

AI & Generative AI: Prompt Engineering, LLM Integration (Gemini, Groq), AI Model Testing, Content Structuring, Generative AI
Programming: Python, JavaScript, Java
Data & ML Tools: Pandas, NumPy, SQL, MongoDB, REST APIs, JSON Data Handling
Web Development: React.js, Node.js, Express.js, Socket.io, HTML5, CSS3, Tailwind CSS
Tools & Platforms: Git, GitHub, Vercel, Render, Postman, Stripe API

EXPERIENCE

Full Stack AI Integration Developer — IFOA (International Flight Operational Academy) *March 2026 – Present*

- Architected a provider-agnostic **LLM abstraction layer** (Gemini primary, Groq/OpenAI-compatible fallback) with automatic failover and token-usage logging to drive AI-generated rule explanations, scenario-based decision trees, and auto-graded quizzes; built an Edge-TTS audio pipeline with SHA-256 token signing and clock-skew correction, plus Mermaid.js concept-flow rendering and Mongoose-cached AI responses.
- Engineered a **JWT/bcrypt-secured MERN service** generating EASA Part ORO.GEN.110(c) and ICAO Doc 10106/9868-compliant PDF certificates via pdf-lib and PDFKit template injection, with Mongoose schemas modeling 12 selectable recurrent-training modules and full participant CRUD.
- Built a **Stripe-integrated subscription billing service** with the Gemini SDK, MJML-templated transactional email pipelines, node-cron-scheduled renewal jobs, ExcelJS reporting exports, and helmet/express-rate-limit hardening; resolved 13+ defects across the payment, renewal, and email subsystems.
- Contributed to a MERN **operations platform** with Socket.io real-time event broadcasting, Multer file-upload handling, express-validator request validation, and role-gated HR/sprint/leave-management modules.

Web Development Intern — Brainpulse, Noida *June 2025 – August 2025*

- Built responsive React.js front-end features that improved UI responsiveness by 20%; automated content-formatting workflows with Python scripts and collaborated in an agile team across multiple client projects.

EDUCATION

B.Tech, Computer Science & Engineering — Dronacharya Group of Institutions (AKTU) | CGPA: 7.2 | 2022 – 2026
St Columbus School — Class XII: 80% (2022) | Class X: 91% (2020)

PERSONAL PROJECTS

Horizn — AI-Integrated Travel Listing Platform · React.js, Node.js, Express.js, MongoDB, Mapbox, Cloudinary [↗ Live](#)

- Designed structured RESTful APIs with Joi validation and role-based authorization middleware for secure content management.
- Integrated Mapbox geocoding and Cloudinary cloud APIs, gaining hands-on experience connecting and testing third-party services and automating data transformation pipelines.
- Implemented Passport.js authentication with MongoDB-persisted sessions, applying production-grade auth patterns.

NexChat — Real-Time Messaging Application · React.js, Node.js, Express.js, Socket.io, MongoDB, JWT [↗ Live](#)

- Built an event-driven WebSocket architecture enabling sub-50ms real-time message delivery.
- Implemented JWT authentication with bcrypt hashing and MongoDB persistence for secure, stateful session logging.

WorldAtlas — Structured Data Explorer · React.js, REST Countries API [↗ Live](#)

- Built a React SPA fetching and filtering structured data for 250+ countries with multi-criteria search by region, population, and area.
- Implemented dynamic routing and drill-down detail pages handling nested API responses.

ACHIEVEMENTS & CERTIFICATIONS

Honors: Smart India Hackathon (SIH) Qualifier · Microsoft Imagine Cup Qualifier · 1st Prize, Technical Quiz · 1st Prize, Circuit Design · 2nd Prize, Hackathon · Bootcamp Selectee

Certifications: Full-Stack Web Development · Data Structures & Algorithms · Cisco Cybersecurity

Currently Pursuing: Prompt Engineering for Generative AI · Machine Learning Fundamentals (Python)